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Dynamic Validation of CNN-Based Surrogate Models for Inverter-Based Resources in Open-Source Solvers

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ABSTRACT Traditionally, distribution system planning has focused on steady-state analyses, with limited consideration of dynamic behavior. However, as large or medium-scale inverter-based resources (IBRs), particularly grid-following (GFL) inverters in commercial or industry buildings, become more prevalent, understanding their dynamic impact is essential for grid planning and operation. This article presents an innovative deep-learning (DL)-approach using convolutional neural networks technique to model the GFL inverters. Developed from real grid-tied commercial IBR transient data, these dynamic DL models overcome proprietary constraints by requiring minimal knowledge of internal converter physics while maintaining high accuracy and flexibility. To demonstrate their applicability, the models were incorporated into GridLAB-D, an open-source, three-phase distribution analysis tool. This integration enables dynamic simulations of large-scale distribution networks with high IBR penetration stability analysis. Rigorous testing and validation, aligned with industry standards, confirmed the reliability and efficiency of this approach, paving the way for enhanced planning and operational assessments of modern power systems.

INDEX TERMS Deep-learning, grid-following, industry standard, inverter-based resources, open-source.

I. INTRODUCTION

INVERTER-BASED resources (IBRs) are being deployed across bulk power and distribution systems at unprecedented scale. By 2050, global IBR capacity is projected to reach 14,500 GW, supplying nearly three-fifths of total electricity generation [1]; in the U.S., IBRs are expected to provide roughly 82% of electricity consumption [2].

Historically, distribution planning has emphasized steady-state and quasi-steady-state analysis, owing to strong substation voltage support and low IBR penetration. With rapid adoption of grid-connected inverters, however, their dynamic behavior has become increasingly important [3]. Although IBRs can enhance grid controllability, their variable and software-driven nature introduces new operational challenges. Recent events, such as the Odessa disturbances, revealed discrepancies between proprietary models and field behavior, underscoring the need for accurate dynamic models and continuous performance monitoring to maintain reliability and resilience [4]. As grid conditions become more inverter-dominated, precise dynamic models are essential for interconnection studies, planning, and real-time operations.

Dynamic modeling of IBRs is typically performed using EMT models [16] or electromechanical models [17]. Both can capture inverter and system dynamics for stability and reliability studies [18], but each has limitations. EMT models provide high fidelity for individual devices yet become impractical for large networks due to computational burden and the need for detailed, configuration-specific parameters [19]. Such information is often proprietary [16], forcing grid operators to rely on generic models that require repeated calibration and validation before use [12].

TABLE 1. Comparison of data-driven modeling studies for grid-following inverters.

Reference	Scenario / data	Modeling approach	Limitation relative to this work
Poudel et al. [5]	Single-phase GFL; measured time-series	Classical Black-Box (BB) identification from terminal variables	Single-phase only; no integration in open-source solvers; validated on isolated tests.
Rojas [6], Krishnamoorthy [7], Kaufhold [8]	Single-phase converters; simulation data	Deep NN–based BB emulation	Simulation-only, single-phase; no feeder-level or network validation.
Li [9], Zhang [10]	Converter admittance data (freq-domain)	Neural mapping of frequency responses (impedance / admittance Identification (ID))	Small-signal focus; no time-domain dynamics or network-level use.
Fan et al. [11]	Utility IBRs; lab + field data	dq-admittance and reduced-order BB/gray models	Emphasis on small-signal/SSR behavior; not embedded in open-source solvers.
Mahapatra et al. [12]	Three-phase inverter digital twin; sim + limited measurements	BB digital twin for operator use	Mainly simulation-based; limited feeder testing; no open-source solver deployment.
Sparse identification (SINDy) and deep symbolic regression (DSR) studies [13], [14]	GFL converters; synthetic EMT / SMIB data	Sparse ID (SINDy) and symbolic regression	Synthetic single-device setups; high computational cost; no multi-vendor or network validation.
Subedi et al. [15]	Commercial GFL inverter; three-phase distribution solver	convolutional neural network (CNN)-based dynamic model in open-source solver	Single-vendor only; limited operating-condition diversity.
Current Work	Two-vendor commercial GFL inverters; field + simulation data	CNN models designed for plug-in use in open-source three-phase solvers	Multi-vendor, field-validated dynamic models with full network-level integration and cross-validation.

Electromechanical models, while adequate for transmission-level analysis, assume balanced conditions and are not suitable for three-phase distribution-level studies [19].

Given the computational demands and confidentiality barriers of EMT models, and the structural assumptions inherent in electromechanical models, machine-learning-based data-driven approaches have emerged as a compelling alternative [20], [21]. These methods can capture IBR dynamics directly from empirical data, support large-scale simulations with high penetration of inverters, and iteratively refine black-box behavior to reduce modeling errors where accuracy is most critical. Several data-driven approaches have been developed to model the dynamics of GFL inverters [22]. These methods learn input–output dependencies without replicating internal control structures, making them suitable when physics-based modeling is difficult. Black-box techniques span classical system identification [5] to deep-learning (DL) models [20].

Most prior work targets single-phase converters [6], [7], [8] or focuses on impedance/admittance identification using simulation-derived frequency-domain data [9], [10]. Measurement-based digital twins for three-phase inverters have recently been proposed [12], but these remain primarily simulation-oriented and lack rigorous real-feeder validation. Subedi et al. [15] introduced the first CNN-based dynamic inverter model implemented within an open-source three-phase distribution solver. Despite these advances, most DL surrogates rely on synthetic RMS/EMT data and are evaluated under limited operating conditions. As a result, they often miss vendor-specific control behaviors, show weak cross-hardware generalization, and are rarely benchmarked against industry model-quality requirements. Our earlier work [15] partially addressed these gaps through a proof-of-concept CNN surrogate for a commercial three-phase GFL inverter in GridLAB-D, though limited to one vendor and a narrower range of transients. A detailed comparison with prior GFL modeling studies is provided in Table 1.

This work extends our earlier study in three ways. First, the CNN surrogate is trained on *field measurements* from multiple commercial GFL inverters, capturing vendor-specific nonlinearities not represented in generic RMS or EMT models. Second, the surrogate is integrated into GridLAB-D as a *Norton-equivalent current source* that maps three-phase terminal voltages to currents within the simulator's quasi-dynamic phasor framework. Third, the model is validated using *ERCOT-aligned* tests, including LVRT, HVRT, frequency steps, phase-angle jumps, DC-input changes, and system-strength variations [23]. The main contributions of this paper are as follows:

- A CNN-based black-box model trained on transient field data from multiple commercial GFL inverters.
- Integration of the surrogate into open-source GridLAB-D via a Norton-equivalent current-source interface.
- Validation under ERCOT-defined disturbances, including LVRT, HVRT, frequency steps, phase jumps, and system-strength variations.

The rest of the paper is organized as follows. Section II presents the fundamentals of the CNN-based DL architecture. Section III describes the system model and methodology, and results are presented in Section IV. Section V concludes the study and outlines future directions.

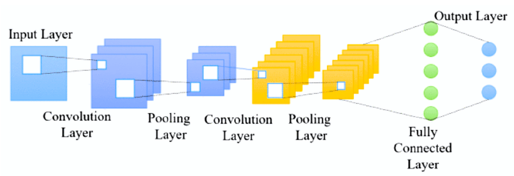

FIGURE 1. Basic CNN architecture.

II. DEEP-LEARNING-BASED DYNAMIC MODELING OF COMMERCIAL GRID-TIED IBRS

A CNN is a feed-forward DL model that uses learnable filters to extract features from data with a grid-like structure. CNNs are widely used in image recognition and typically consist of convolutional layers, pooling layers, and fully connected layers, as illustrated in Fig. 1. Convolutional layers slide filters over the input to form feature maps, pooling layers reduce dimensionality, and fully connected layers perform regression or classification. All parameters are learned via back-propagation during supervised training [24], [25].

In this work, the inverter surrogate adopts a standard one-dimensional CNN backbone with stacked convolution, activation, and fully connected layers. The architecture itself is conventional; the specialization comes from the selected input/output phasor variables, the sequence length, and key hyperparameters such as filter count and regularization. The CNN receives short time windows of three-phase voltage phasors and outputs the corresponding current phasors, learning a transient input–output mapping rather than a static algebraic relationship [25].

This structure is well suited for dynamic studies: the temporal receptive field of the 1D convolutions captures the effects of internalhase-locked loop (PLL), current-control, and outer-loop dynamics over the window, effectively approximating the solution operator of the underlying differential equations. Traditional inverter models based on detailed differential–algebraic equations require proprietary control details and often yield stiff systems that are expensive to integrate. In contrast, the CNN surrogate is trained directly on measured transient responses, implicitly learns fast nonlinear behavior, and interfaces with the network as a Norton-equivalent current source at each simulation step. This offers a practical balance between dynamic fidelity and ease of deployment for phasor-domain transient simulations.

While convolutional layers extract features from input vectors or matrices, pooling layers reduce their spatial or temporal dimensions. During supervised training, convolutional filters slide across the input to produce feature maps, and all filter weights are learned through back-propagation. Fully connected layers in the final stages perform classification or regression, yielding probability outputs for classification tasks or real-valued outputs for regression problems. The operation of a convolutional layer C_i is expressed as [26]:

$$C_i = f\left(\sum_k k_{ij} x_i\right), \tag{1}$$

where x_i is the input feature, k_{ij} is the convolutional kernel, and $f(\cdot)$ is the nonlinear activation function.

The max-pooling operation applied to a local neighborhood of size $j \times k$ in an input map X is given by [26]:

$$h_{ijk} = \max(X_{i,j+p,k+q}), \tag{2}$$

where p and q denote the vertical and horizontal offsets within the pooling window.

A fully connected layer Ψ is defined as [26]:

$$\Psi = f(W_\Psi y + b), \tag{3}$$

where W_Ψ and b are the weight matrix and bias, and y is the layer input.

A. PHASOR EXTRACTION USING FAST FOURIER TRANSFORM

Point-on-wave voltage and current measurements from commercial GFL inverters were converted into phasor quantities using a short-time fast Fourier transform (FFT) approach implemented in MATLAB [27]. For each waveform the discrete FT was computed using the `fft` function. Let f_s denote the sampling frequency and f_0 the nominal system frequency (60 Hz). The fundamental phasor was obtained from the FFT bin as

$$k = \left\lfloor \frac{f_0 N}{f_s} \right\rceil,$$

and the complex coefficient $X[k]$ was scaled by $2/N$ to yield the magnitude and phase of the fundamental component. This method provides a consistent and noise-robust estimate of the instantaneous phasor suitable for dynamic studies.

B. INPUT OUTPUT

In this work, the CNN-based DL surrogate is formulated as a mapping from three-phase terminal voltage phasors to three-phase current injections. The inputs to the model are the three-phase voltage magnitudes $(V_{a_{\text{mag}}}, V_{b_{\text{mag}}}, V_{c_{\text{mag}}})$ and their corresponding phase angles $(V_{a_{\text{ang}}}, V_{b_{\text{ang}}}, V_{c_{\text{ang}}})$, while the outputs are the three-phase current magnitudes $(I_{a_{\text{mag}}}, I_{b_{\text{mag}}}, I_{c_{\text{mag}}})$ and associated phase angles $(I_{a_{\text{ang}}}, I_{b_{\text{ang}}}, I_{c_{\text{ang}}})$. This $V_{abc} \mapsto I_{abc}$ formulation makes the surrogate behave as a Norton-equivalent device, consistent with quasi-dynamic and phasor-domain solvers such as GridLAB-D, which operate on bus voltage phasors and current injections. It also reflects how GFL inverters are controlled in practice, since PLL and current-control loops act on terminal voltages, and it naturally captures both balanced and unbalanced fundamental-frequency behavior. The dataset is divided into training and validation subsets, where the training set is used to fit the model and the validation set is used to assess its generalization performance.

C. LOSS FUNCTION

The loss function measures the difference between the model's predictions and the actual target values during training. In this study, the Mean Absolute Error (MAE) is utilized as the loss function [28]. MAE calculates the mean of the absolute differences between the predicted and true values as follows:

$$J(\theta)(\text{MAE}) = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|, \tag{4}$$

where n refers to the total number of samples, y_i represents the actual value for the i^{th} sample, and $\hat{y}_i$ corresponds to the predicted value for the same sample. Smaller MAE values indicate improved model performance.

D. OPTIMIZATION ALGORITHM

Training a DL model involves solving an optimization problem that minimizes the loss by updating network parameters. In this work, gradient-based optimization is used, and the Adam optimizer is selected for its efficiency and robustness. Adam combines ideas from AdaGrad and RMSProp [29] by maintaining exponential moving averages of the first and second moments of the gradients. Its update rules are:

$$\begin{aligned} m &= \beta_1 m + (1-\beta_1)\nabla J(\theta),\\ v &= \beta_2 v + (1-\beta_2)(\nabla J(\theta))^2,\\ \theta &= \theta - \alpha\frac{m}{\sqrt{v}+\epsilon}, \end{aligned} \tag{5}$$

where θ denotes the model parameters, $J(\theta)$ the loss function, and $\nabla J(\theta)$ its gradient. The learning rate α controls the step size, β_1 and β_2 are decay rates for the moment estimates, m and v are the first and second moments, and ϵ (10^{-8}) prevents division by zero.

E. ACTIVATION FUNCTION

Activation functions add non-linearity to DL models, allowing them to capture complex relationships between inputs and outputs. These functions transform the output of each neuron in the network, defining the neuron's response to a given input. Commonly used activation functions include the Rectified Linear Unit (ReLU), Leaky ReLU, and Exponential Linear Unit (ELU) [24], [30]. This study adopts the ReLU activation function due to its computational efficiency in training neural networks. The ReLU function is defined as:

$$ReLU(x) = \max(0, x), \quad x \in \mathbb{R}. \tag{6}$$

F. MODEL ACCURACY

To evaluate model performance during training and validation, evaluation metrics are employed. For regression problems, the MAE serves as the primary accuracy metric, quantifying the deviation of predictions from actual values.

G. PREPROCESSING DATASET

The model operates in the phasor domain, consistent with its implementation in the distribution solver GridLAB-D. Three-phase voltage and current measurements are decomposed into magnitude and phase-angle components. The raw data then undergo several preprocessing steps to improve DL model accuracy, including removal of start-up and corrupted data, downsampling, scaling, and dataset partitioning. Extraneous start-up segments are first removed, and measurement or conversion errors are corrected. Data from multiple test conditions (frequency jumps, voltage ride-through, and DC input changes) are then concatenated and downsampled to reduce computational cost. Finally, all inputs and outputs are normalized and divided into training and validation sets. The detailed preprocessing and dataset construction procedure follows the methodology outlined in [15].

H. PERFORMANCE EVALUATION METRICS

Data-driven models derive system dynamics by creating mathematical representations based on empirical data without requiring explicit system knowledge [31]. The DL model's performance, when integrated into the open-source software, is assessed using normalized Root Mean Square Error (NRMSE):

$$NRMSE = \frac{||y(t) - \hat{y}(t)||}{||y(t) - \text{mean}(y(t))||}, \tag{7}$$

where $\hat{y}(t)$ denotes the estimated values, and $y(t)$ represents the actual values.

III. SIMULATION SETUP: TRAINING AND VALIDATION

A. OVERVIEW

Fig. 2 shows an overview of the proposed DL model of IBR (a) data extraction, (b) pre-processing, (c) training, (d) integration with an open-source solver (GridLAB-D), and (e) testing in accordance with industry standards outlined in the ERCOT model quality test manual. This framework ensures accuracy and applicability for modeling IBRs.

B. EXPERIMENTAL SETUP FOR DATA EXTRACTION

The process begins with experimental data collection from two distinct commercial inverters, referred to as IBR-1 and IBR-2 as shown in Fig. 2–(a). The first dataset comes from a grid-tied, level-5, 1 MW GFL inverter tested under three scenarios: voltage ride-through, frequency jumps, and DC input variations. Three scenarios are obtained under balanced system conditions by switching the circuit breakers (CB_2) to create a fault, varying line impedance (R_2 and L_2) to simulate ride-through conditions, and directly changing the DC source modeled in RSCAD for DC input variations. During each transient event, three-phase voltage, current, and DC input are recorded at a 1.2 ms sampling rate to train the DL model.

Similarly, the second dataset is obtained from a laboratory setup featuring a 125 kVA GFL PV inverter connected to a 540 kVA grid simulator. This setup demonstrates the applicability of the DL model to another commercial IBR product. Two tests were conducted under balance system conditions: phase jump, and frequency jump. The three-phase voltage and current are recorded at a 0.2 ms sampling rate during each transient event to train the DL model.

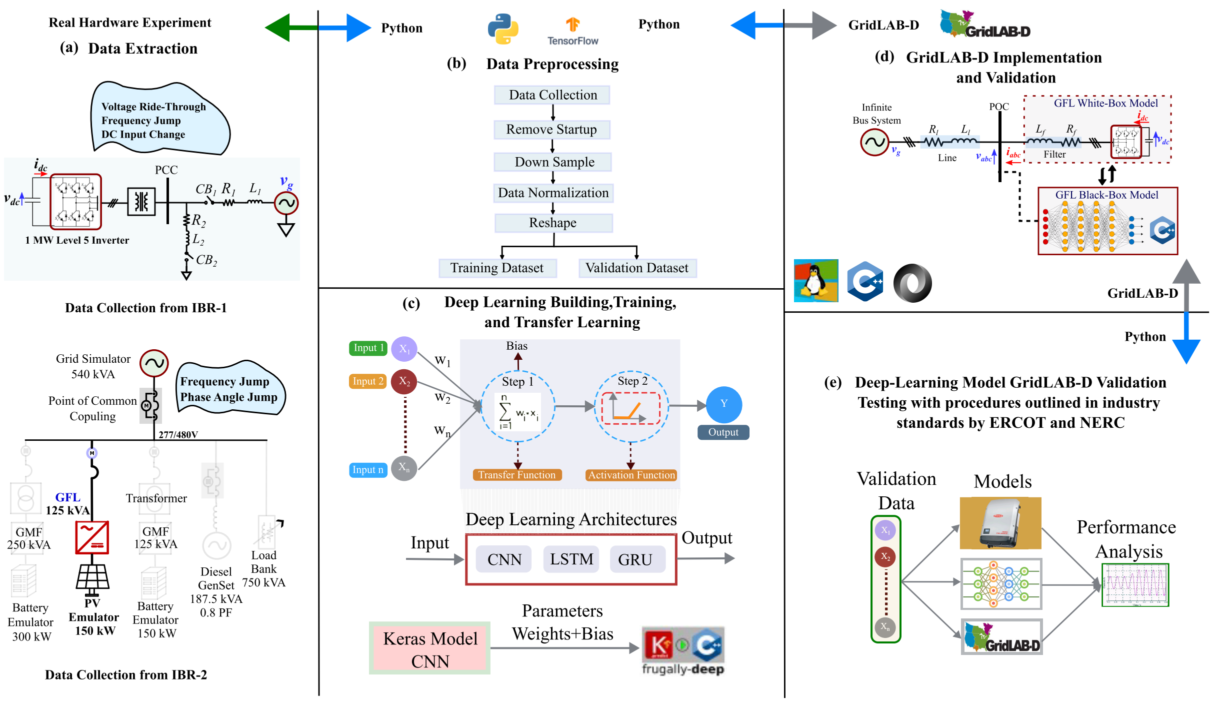

FIGURE 2. Overview of the proposed DL model of IBR training, integration, and testing.

C. DL MODEL ARCHITECTURE AND TRAINING

The collected point-on-wave time-series data were converted into phasor quantities using an FFT-based method. Consecutive phasor estimates were generated with 50% overlap to improve temporal resolution. Anti-alias filtering was applied before downsampling, after which the phasor magnitudes and angles were normalized and uniformly downsampled by a factor of 10 to reduce computational cost and improve numerical stability. The resulting phasors were arranged into sliding windows of five consecutive samples for CNN training, providing a consistent representation of transient behavior.

The processed voltage and current magnitudes and angles are then reshaped into a tensor structure for training and validation, as shown in Fig. 2(b). Several DL architectures were evaluated, including basic RNN, LSTM, GRU, Bi-LSTM, and Bi-GRU networks. Their performance in terms of training accuracy, validation accuracy, and training time is summarized in Fig. 6.

Among the tested models, the CNN achieved the best accuracy–runtime trade-off, with the highest validation accuracy, a small train–validation gap, and competitive training time, indicating strong generalization at low computational cost. Other architectures either trained more slowly or showed larger generalization gaps, suggesting higher risk of overfitting. Based on these results, the CNN was selected for the closed-loop simulations.

The CNN surrogate model was trained and validated on datasets that capture a range of IBR operating conditions with varying control parameters, ensuring that the model learns from diverse system dynamics. The CNN model takes six inputs i.e., three-phase voltage magnitudes and angles and six outputs i.e., three-phase current magnitudes and angles injected by IBR, as shown in Fig. 4. The CNN model is implemented using Keras Sequential API in Python with TensorFlow. The multi-input, multi-output model has a spatial input tensor dimension of $5 \times 6 \times 1$, where the height 5 represents five samples of data, the width 6 corresponds to three-phase voltage magnitude and angle, and the channel of 1 represents a single-channel input. The hidden layers include a combination of convectional (or convolutional) layers, max pooling, flatten, culminating in a fully connected layer. The convolutional layer consist of $3 \times 3 \times 64$ neurons, while the dense layer consists of 14 hidden neurons. The *ReLU* activation function is utilized to learn the complex patterns with a batch size of 32 and total trainable parameters of 61,934. Moreover, the Adam optimizer is used for efficient and adaptive learning rate properties. The model's accuracy is accessed using MAE over multiple epochs on the training data. The summary of the settings used of the training is summarized in Table. 2.

For transfer learning from IBR1 to IBR2, the convolutional layer and the first dense layer were frozen to retain the feature representations learned from IBR1, and only the final dense layer was fine-tuned using IBR2 data. Fine-tuning was performed with the Adam optimizer at the default learning rate (10^{-3}), without learning-rate scheduling or early stopping. Although evaluation across all IBR control strategies is beyond the present scope, the framework readily accommodates additional datasets and supports efficient

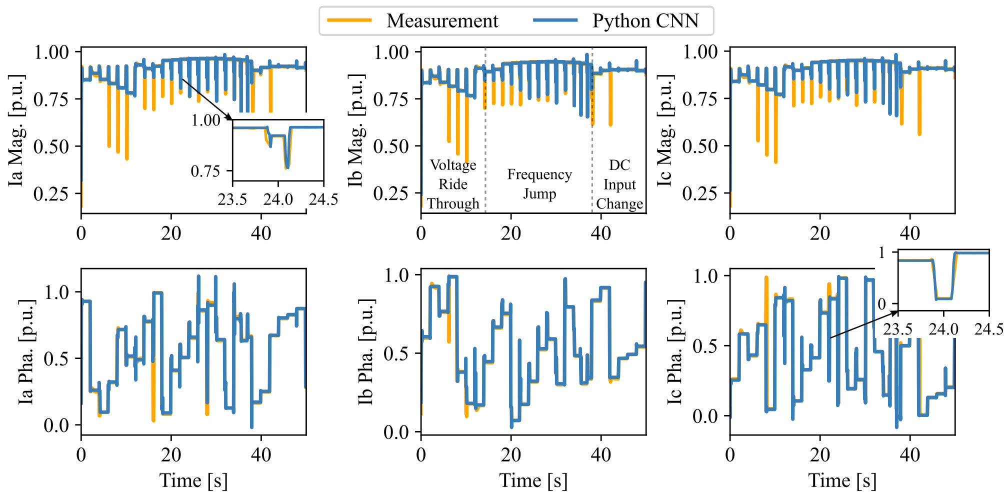

FIGURE 3. Three-phase output current magnitudes and phase angles measurement from the commercial inverter (IBR-1) versus the outputs from the DL CNN model with the training data from IBR-1.

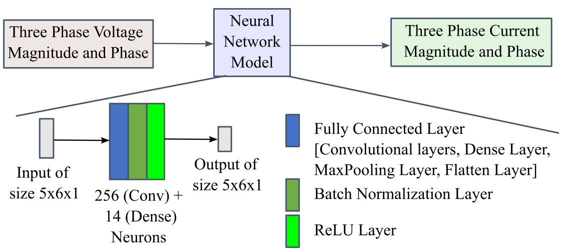

FIGURE 4. The structure of CNN model representing the IBR-1 and IBR-2.

TABLE 2. Data preprocessing and CNN training setup.

Item	Setting
Raw sampling interval	IBR1: 1.2 ms, IBR2: 0.2 ms
Downsampling factor	10
Effective timestep	IBR1: 12 ms, IBR2: 2 ms
Sliding window, Seed	5 samples, 42
Input features	$\lvert V_{a,b,c}\rvert$, $\angle V_{a,b,c}$
Output features	$\lvert I_{a,b,c}\rvert$, $\angle I_{a,b,c}$
Input/Output tensor	$5 \times 6 \times 1$
CNN model	Conv2D(64, 3×3) → MaxPool(2,2) → Flatten → Dense(14) → Dense(6)
Activation, Loss, Optimizer	ReLU, MAE, Adam
Batch size, Epochs	32, 500
Split	80/20 (chronological)
Hardware	Dell OptiPlex 7090, i7, 16 GB RAM

retraining, enabling adaptation to new operating conditions and use in optimization or controller-design studies.

D. CASE STUDY-I: COMMERCIAL IBR-1

In case study-I, the DL model is trained to represent the dynamics of IBR-1 under three operating scenarios: voltage ride-through, frequency jump, and DC input change as shown in Fig. 2–(c). Fig. 3 shows the performance of the trained model in representing the dynamics of IBR-1 using training data. Fig. 7 shows the loss during the training and the validation over the number of epochs. The results indicate that the trained DL model accurately represents the wide dynamic operating range of the commercial IBR with minimal error, achieving NRMSE of below 4.5%.

E. CASE STUDY-II: COMMERCIAL IBR-2

The pre-trained model using data from IBR-1 is retrained and finetuned with new data from IBR-2. This process enables the model to adapt the knowledge learned from IBR-1 to a new dataset, which contains frequency and phase-angle jump scenarios. Then the new trained model is used to validate against the additional data from IBR-2 to assess its generalization capability across different IBR dynamics. Fig. 5 shows the validation performance of the tuned model with new data from the new commercial IBR. The results confirm that the model accurately represents the dynamics of the new commercial IBR product under various operating scenarios, with a minimal validation NRMSE below 5%.

F. OPEN-SOURCE SOLVER IMPLEMENTATION

The CNN-based surrogate is developed and trained in Python using Keras/TensorFlow. After training, the network architecture and weights are exported to a lightweight JSON format using the Frugally-deep `keras_export` tool [32]. On the C++ side, the Frugally-deep library loads this JSON file and provides a compiled inference engine that is linked into a custom GridLAB-D module. This workflow enables deployment on non-GPU platforms without requiring a Python or TensorFlow runtime, as shown in Fig. 2(c)–(d).

To benchmark performance, the surrogate was first evaluated in an infinite-bus test system supplying a single GFL inverter, as shown in Fig. 2(d). The detailed model of the GFL inverter used in our simulations is based on the

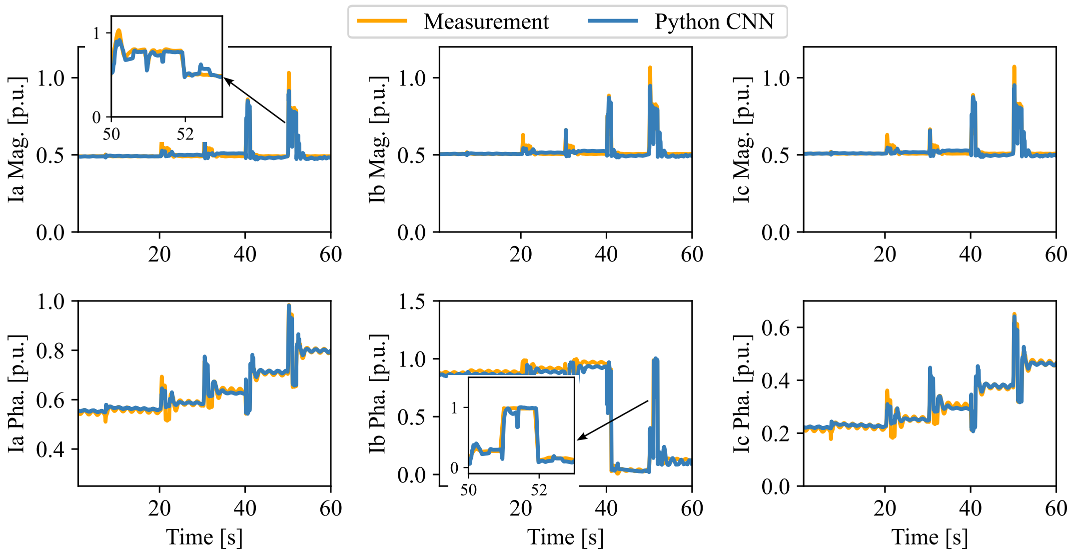

FIGURE 5. Three-phase output current magnitudes and phase angles measurement from the commercial inverter (IBR-2) versus the outputs from the DL CNN model with the validation data from IBR-2.

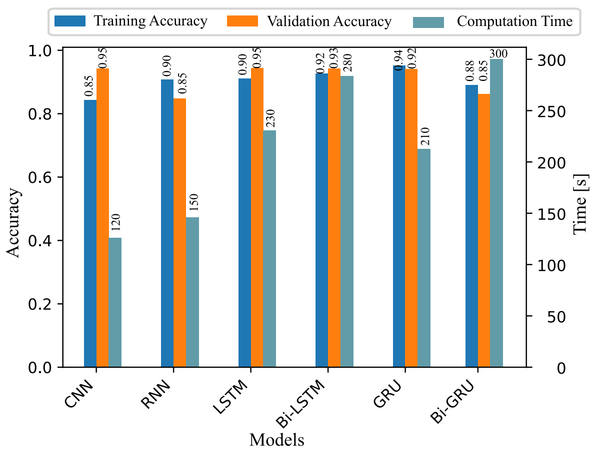

FIGURE 6. Training accuracy, validation accuracy, and computation time for different DL models.

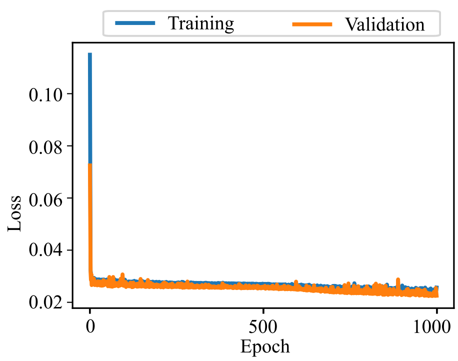

FIGURE 7. Training and validation loss over number of epoch for IBR-1.

formulation presented in Du et al. [19]. All simulations used GridLAB-D in Δ-mode with a fixed timestep of 2 ms over a 40 s window (approximately 20,000 iterations). Experiments were conducted on a Dell OptiPlex 7090 desktop (Intel Core i7 vPro, 16 GB RAM). Under this configuration, the detailed physics-based GFL model required 209 s of computational time.

The detailed GFL model was then replaced by the trained surrogate, which is integrated directly into GridLAB-D's Δ-mode phasor solver. At each 2 ms timestep, GridLAB-D provides the inverter's terminal three-phase voltages; a five-sample sliding window is formed, converted to magnitude–angle features, and passed to the CNN. The network outputs the corresponding three-phase current phasors, which are injected as a Norton-equivalent source. This interface allows the surrogate to emulate the dynamic behavior of a GFL inverter—including PLL and current-control responses—while remaining fully compatible with GridLAB-D's phasor-domain framework. With the surrogate in place, the total simulation time was reduced to 198 s. Although the speedup is modest for a single-IBR case, the surrogate preserves sub-millisecond resolution while reducing per-step computational cost, and its scalability advantages are expected to grow for larger IBR-rich systems.

The surrogate operates as a self-contained module in GridLAB-D, using only phasor-level voltage inputs and requiring no external co-simulation. New device-specific models can be produced from field data through simple preprocessing and transfer learning, without manual neural-network tuning, and integrated in the same way. The implementation follows the GridLAB-D Black-Box IBR Models interface, with usage instructions documented on the GridLAB-D wiki [33].

IV. RESULTS AND ANALYSIS

A. PERFORMANCE COMPARISON

Real data from a commercial GFL inverter are used to train and validate the surrogate. The three-phase voltage

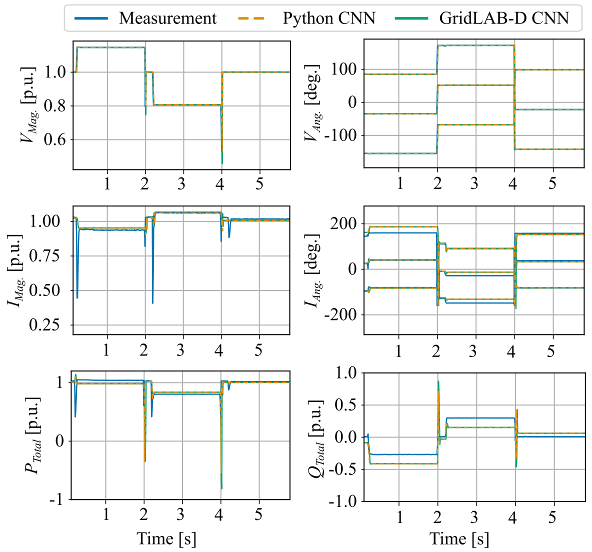

FIGURE 8. Comparison of three-phase voltage, current, and active /reactive power from commercial IBR measurements, the Python CNN model, and the CNN integrated in GridLAB-D.

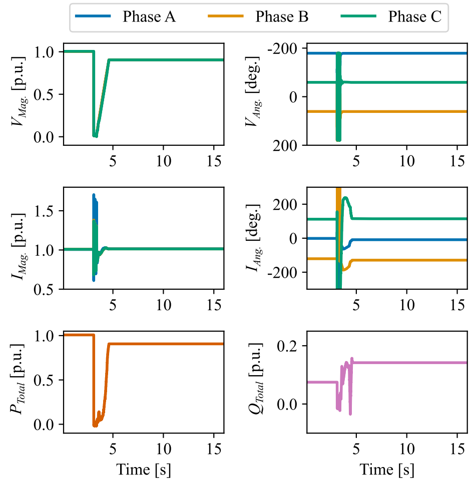

FIGURE 9. Outputs of the proposed DL model during LVRT.

magnitudes and angles extracted from the point-on-wave recordings serve as inputs to the DL model, while the corresponding current magnitudes and angles form the outputs. After integrating the trained model into the open-source solver, its behavior is compared against both the Python implementation and the real measurements, as illustrated in Fig. 2–(e). This setup establishes a consistent input–output framework and enables a direct assessment of the surrogate's fidelity across measurement, Python inference, and GridLAB-D integration. Results show that the integrated model accurately tracks the Python CNN-BBM and captures the dynamics observed in the real measurements. Fig. 8 compares the three-phase voltage and current phasors and the active/reactive power from the commercial IBR, the Python CNN-BBM, and the GridLAB-D implementation. For the current magnitude and angle, the Python model achieves a training NRMSE of 4.0% and a validation NRMSE of 4.5% against the measured field data. The validation error for active and reactive power is slightly higher at approximately 5.2%, reflecting the additional sensitivity of power calculations to small phasor deviations. These discrepancies may primarily arise from measurement noise, the point-on-wave to phasor conversion, and normalization. After export to JSON/Keras and execution through the Frugally-deep C++ engine, the GridLAB-D implementation matches the Python CNN within 0.5% across all channels, confirming that the surrogate preserves its fidelity when deployed inside the open-source solver.

The surrogate model is trained on normalized field data and is therefore valid only within the voltage and angle ranges represented in that dataset. In the absence of explicit runtime guardrails, these training-domain limits naturally define where the BBM can be trusted. The BBM is appropriate for steady-state and moderately disturbed conditions, but it is not a replacement for detailed EMT models. Researchers should revert to EMT or physics-based phasor simulations whenever operating conditions fall outside the trained range for example; in very weak grids (SCR ≤ 3 or below), under large unbalances, or for studies requiring switching-level fidelity or detailed control-loop analysis.

B. QUALITY ASSESSMENT OF DYNAMIC DL MODELS

To further evaluate the DL model, representing GFL IBR dynamics, within an open-source distribution software, testing procedures are derived from the model validation test procedures outlined in the dynamic working group procedure manual (revision 22) from ERCOT [23]. The DL model is evaluated under various operating conditions, including system strength variations, low- and high- voltage ride-through scenarios, small frequency disturbances, phase angle jumps, and both balanced and unbalanced faults.

1) VOLTAGE DISTURBANCE TEST

Fig. 9 presents the output of the GridLAB-D DL model during a low voltage ride-through (LVRT) test, where the voltage at the point of interconnection (POI) is reduced from 1.0 p.u. to 0.9 p.u. As expected, when the POI voltage dropped to zero, both active and reactive power at the POI dropped to zero. The active power recovered shortly after the voltage ramped up from zero. When the POI voltage stabilized at 0.9 p.u., a small reduction in active power was observed due to the reactive power injection. Overall, the DL model remained stable and closely mirroring the dynamic behavior of a real IBR.

Fig. 10 shows the output of the GridLAB-D CNN-BBM during high voltage ride-through (HVRT) test, where the POI voltage increased from 1.0 p.u. to 1.1 p.u. During the HVRT event, active power output remained stable. However, when the voltage further reached 1.2 p.u., the model exhibited a sharp absorption of reactive power in response to the voltage change. After a few cycles, the reactive power leveled off and remained at a fixed value.

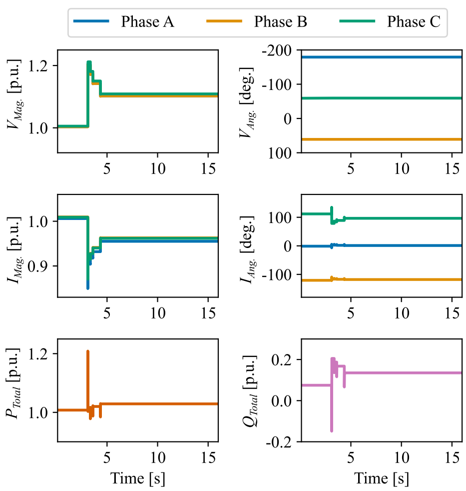

FIGURE 10. Outputs of the proposed DL model during HVRT.

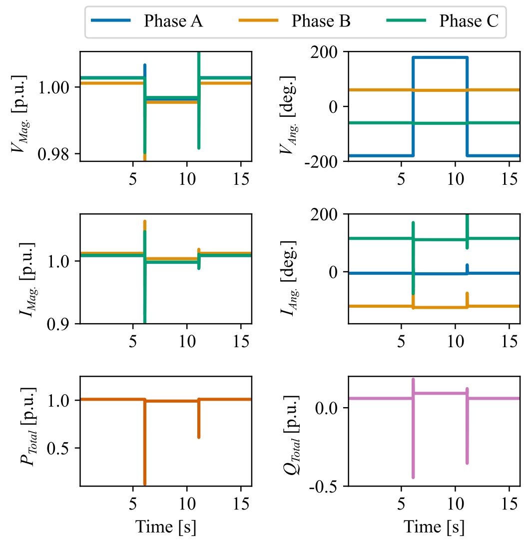

FIGURE 11. Outputs of the proposed DL model during strong system strength test with SCR of 5.

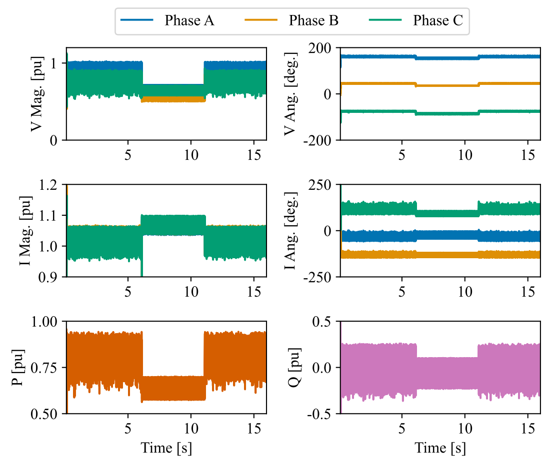

FIGURE 12. Outputs of the proposed DL model during during weak system strength test with SCR of 3.

2) SYSTEM STRENGTH TEST

The short-circuit ratio (SCR) quantifies grid strength at the POI as the ratio of the Thevenin short-circuit strength to the inverter's rated power. A high SCR indicates a stiff grid with stable PCC voltage, while a low SCR reflects a weaker grid where the PCC voltage becomes sensitive to inverter current disturbances. SCR is especially critical in modern grids with high penetration of GFL IBRs, since their PLL-based synchronization and current-control performance depend strongly on grid strength.

The system-strength evaluation in this study follows the IBR model-quality testing guidance used by ERCOT [23]. In this procedure, SCR is varied by adjusting the line reactance between the inverter and the POI, and the model is expected to remain stable for SCR values of 3 and above. Following these guidelines, SCR levels of 5 and 3 were tested, and a small 2–3% voltage disturbance was applied at the POI to excite the GFL dynamics—consistent with the "small-voltage disturbance test" defined in the model-quality framework [23].

When the grid is strong (SCR $\geq$ 3), the CNN-based surrogate produces stable and well-damped responses with negligible voltage disturbance as shown in Fig. 11. As the grid weakens, the higher grid impedance makes the PCC voltage "soft" and highly sensitive to injected currents. GFL inverters are particularly vulnerable in this regime because their PLL relies on the disturbed PCC voltage to estimate the grid angle [34]. This creates a feedback loop in which small current perturbations distort the voltage and subsequently affect the PLL and current-control loops as shown in Fig. 12. This interaction between grid impedance and GFL control dynamics is known to produce 2–10 Hz oscillations and, in severe cases, loss of synchronism [35], [36]. These oscillations arise from the underlying physics and are also observed in detailed EMT simulations, confirming that they are not artifacts of the surrogate model.

As SCR decreases to approximately 3 (Fig. 12), both the EMT model and the BBM exhibit more pronounced oscillations in voltages, currents, and active/reactive power (P/Q). These results are consistent with system-strength test expectations and demonstrate appropriate weak-grid behavior. While the EMT model remains the high-fidelity reference, the BBM shows slightly reduced damping and minor phase deviations, indicating that this operating point lies near the edge of the surrogate's validated training range.

3) SMALL FREQUENCY DISTURBANCE TEST

To evaluate frequency sensitivity, the integrated DL model in GridLAB-D is subjected to a 0.05 Hz step change from the nominal 60 Hz frequency. Fig. 13 shows the GFL inverter's response to this perturbation. While the DL model remained stable, small oscillations were observed, which may be attributed to limited training data. Expanding the dataset with more comprehensive commercial inverter data would likely improve model accuracy and prediction reliability.

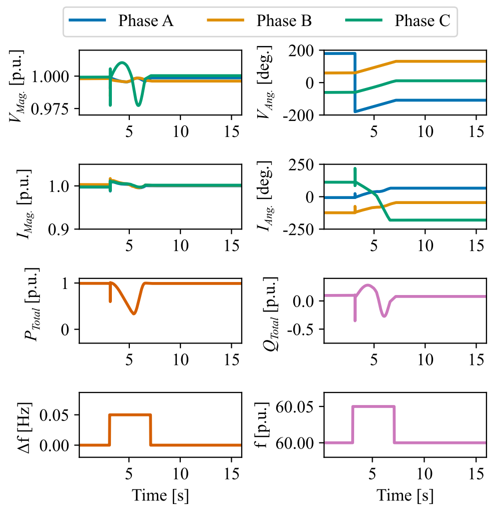

FIGURE 13. Outputs of the proposed DL model during small frequency disturbance test with 0.05 Hz step change in frequency.

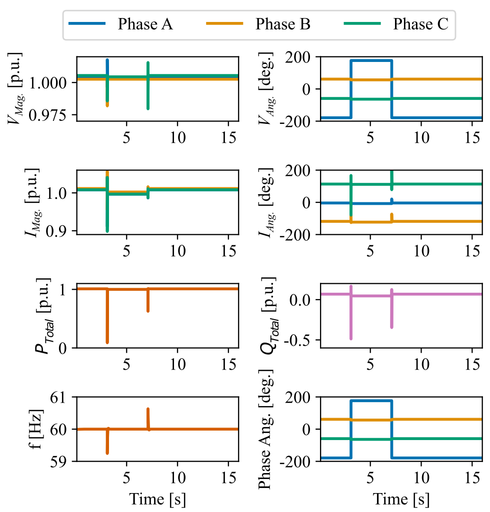

FIGURE 14. Outputs of the proposed DL model during negative 5-degree phase angle jump in the grid voltage angle to emulate small disturbance.

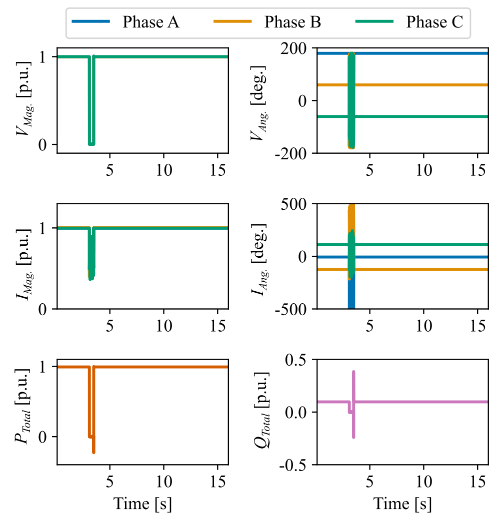

FIGURE 15. Outputs of the proposed DL model during three-phase to ground balanced fault.

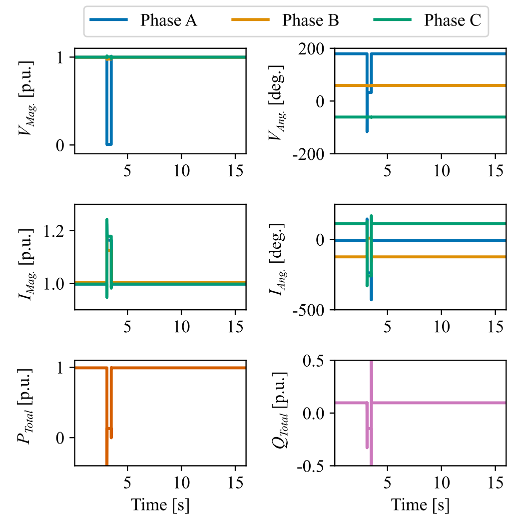

FIGURE 16. Outputs of the proposed DL model during single-line to ground unbalanced fault.

4) PHASE ANGLE JUMP

A negative 5-degree instantaneous phase angle jump is applied to the system, with the response shown in Fig. 14. The system successfully returned to its original steady-state conditions after the disturbance. The results indicate that the DL model of GFL inverter correctly captures phase jump dynamics and remain stable post-disturbance.

5) BALANCE AND UNBALANCE FAULT

To evaluate the fault response of the DL model, it is integrated into an infinite bus system within GridLAB-D, where both balanced and unbalanced faults are introduced. In the balanced fault test, a three-phase-to-ground fault is applied, causing the voltage in all three phases to drop to 0 p.u. The system successfully recovers to its initial steady-state conditions once the fault is cleared, as shown in Fig. 15.

In the unbalanced fault test, a single-phase-to-ground fault is introduced, where the voltage of Phase A drops to 0 p.u., while the voltages of the other phases remain unchanged. The model exhibits appropriate dynamic behavior, successfully returning to its initial state after the fault is cleared, as illustrated in Fig. 16.

The surrogate model provides a fast and efficient alternative to dynamic simulations, predicting system responses under new conditions without directly solving phasor equations. While some error relative to full simulations is unavoidable, validation across diverse scenarios shows the accuracy to be sufficient for system-level studies. For critical cases near protection or stability limits, the surrogate model is best employed as a rapid screening tool, with detailed simulations reserved for final verification.

V. CONCLUSION AND FUTURE WORKS

In this work, a DL-based dynamic model of a GFL-controlled IBR was developed using a CNN architecture. The surrogate addresses gaps in existing models by providing a balance between accuracy, computational efficiency, and implementation flexibility.

The models were trained using measurements from two commercial IBRs under a range of grid conditions, including voltage and frequency ride-through events. The selected CNN surrogate was then integrated into the open-source distribution simulation platform GridLAB-D to enable dynamic analysis. The integrated model was validated against the industry requirements specified in the ERCOT Model Quality Test Manual, demonstrating compliance with established standards and filling an important gap in the literature.

Simulation results confirm that the CNN-based surrogate replicates the dynamic behavior of GFL IBRs with sufficient accuracy for planning and stability studies. This work demonstrates a proof-of-concept integration of a data-driven IBR surrogate into GridLAB-D. While the present study focuses on a single-IBR implementation, current efforts are focused on extending this framework to large-scale, multi-IBR networks, where the computational benefits of surrogate modeling are expected to be more pronounced. This remains an important direction for future research.

CONFLICT OF INTEREST

None of the authors have a conflict of interest to disclose.

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